

Swetha Rathinavelu Saravanakumar

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EXPERIENCE

ML Research Intern

Sastra University

India

Nov 2024 - April 2025

- Built a Transformer-based anomaly detection model on the Secure Water Treatment dataset, achieving >80% detection accuracy across 51 sensor streams in cyber-physical systems
- Developed ETL pipelines to process multivariate time-series data (11 days, high-frequency), ensuring temporal alignment across network and historian logs
- Applied Self-Organizing Maps (SOM) for dimensionality reduction and clustering, isolating patterns across 41 attack scenarios to improve model interpretability

EDUCATION

University of Maryland - College Park

Master of Science in Applied Machine Learning (GPA: 4.00)

College Park, Maryland

August 2025- Present

- Coursework: Probability and Statistics, Principles of Data Science, Principles of Machine Learning, Optimization Techniques, Computer Systems for Machine Learning, and Algorithms and Data Structures for Machine Learning.

Sastra University

Bachelor's in Computer Science and Engineering (GPA: 3.00)

Tamil Nadu, India

October 2020 - May 2024

- Coursework: Software Engineering, Programming and Data Structures, Cloud Computing, IoT, DBMS.

PROJECT EXPERIENCE

QuantOpt: Multi-Strategy Equity Portfolio Optimization ([Project link](#))

- Built an end-to-end portfolio optimization pipeline comparing **Sharpe-Max (MPT)** vs **Risk Parity**, revealing a **~9% volatility reduction** with balanced risk allocation while maintaining competitive returns.
- Implemented Efficient Frontier via **Monte Carlo simulation (1,000 runs)** and solved constrained optimization using **SLSQP**, incorporating real-world constraints (budget, bounds, non-finite data handling).
- Engineered a robust data pipeline (yfinance + preprocessing) and delivered actionable insights on **risk-return tradeoffs across AAPL, MSFT, GOOGL, TSLA**, highlighting concentration vs stability dynamics.

RAG-Based Fitness Assistant ([Project link](#))

- Built a **Retrieval-Augmented Generation (RAG)** system using **Ollama**, **reducing LLM hallucinations via context-grounded responses from a vector database**.
- Designed a semantic retrieval pipeline with **transformer embeddings** and Top-K similarity search to deliver precise, intent-aware exercise guidance.
- Deployed a **fully Dockerized, modular architecture** with an evaluation framework measuring **retrieval hit rate and response relevance**, ensuring production-ready reliability.

WNBA Win Prediction & Performance Analytics ([Project link](#))

- Developed and benchmarked **Logistic Regression, Decision Trees, and Random Forest classifiers**, achieving a 85.8% test accuracy and a balanced 0.86 F1-score in predicting game outcomes.
- Implemented RFECV (Recursive Feature Elimination with Cross-Validation) to isolate high-impact predictors from 4,500+ game logs, maintaining a 94% training accuracy while monitoring for model generalization.
- Conducted rigorous model assessment using **detailed classification reports**, ensuring high precision (0.86) and recall across both winning and losing classes to provide reliable, actionable sports insights.

Azure Computer Vision with Angular ([Project link](#))

- Built an **Angular-based image recognition system using Microsoft Azure Cognitive Services** Computer Vision API to analyze images for categories, descriptions, tags, faces, brands, and color detection.
- Integrated REST APIs with secure subscription key management and optimized frontend architecture using reusable Angular services and HttpClient.
- Implemented **real-time JSON response visualization** for intelligent image analysis and cloud-based computer vision workflows.

TECHNICAL SKILLS

Languages: Python, SQL (joins, subqueries), C/C++, Java | **Machine Learning models,** scikit-learn, RAG (Ollama, embeddings, vector DBs), Monte Carlo, portfolio optimization (MPT, Risk Parity, SLSQP) | **Data:** Pandas, NumPy, ETL, Spark/PySpark, Kafka, yfinance | **Cloud/DevOps:** Azure (CV API), Docker, Kubernetes, Git, Linux | **Web:** FastAPI, Flask, Streamlit, Angular, REST APIs, JSON | **DB/BI:** MySQL, PostgreSQL, Tableau, Power BI, Excel